

Assignment 2:

Date: 11/8/2026

1. Password History System - fun related

WAP with the following rules: The user should be able to change their password.

The new password must not be the same as any of the last 3 passwords used.

The new password must satisfy all of the following conditions:

- Minimum length of 10 characters.
- Must contain at least one lowercase alphabet [a-z].
- Must contain at least one uppercase alphabet [A-Z].
- Must contain at least one numeric character [0-9].
- Must contain at least one special character from [_ @ \$].

After a successful password change, the system should update the password history.

The user can continue to change passwords, and the system should maintain a history of the last 3 passwords only.

2. Hotel booking calculator - positional, keyword n default

A hotel calculates the final bill based on:

- Number of rooms
- Number of nights
- Room price
- Breakfast charges
- Extra-bed charges
- Discount

Some charges are optional, and most bookings use standard values.

The program should allow a booking to be created using different combinations of supplied information.

Example:

Booking 1:
3 rooms, 2 nights

Booking 2:
3 rooms, 2 nights, breakfast charges changed

Booking 3:
3 rooms, 2 nights, discount changed

3. Employee Salary & Benefits System - *args, **kwargs

A company wants a reusable function to generate an employee's salary report.

Every employee has:

- Employee name
- Basic salary
- Any number of **allowances**, where only the amounts matter
- Any number of **deductions**, where each deduction has a meaningful name such as PF, Tax, Insurance, Loan

Calculate and display:

- Total allowances
- Gross salary
- Total deductions
- Net salary
- Complete deduction details

Example:

Employee: Rahul
Basic Salary: ₹50,000

Allowances:
₹5,000
₹3,000
₹2,000

Deductions:
PF = ₹3,000
Tax = ₹2,000
Insurance = ₹1,000

- 4. Number Group Analyzer - Nested fun, Recursive fun, lambda, *args**
A mathematics application accepts a list of positive integers from the user. The application needs to analyze the numbers and produce a summary.

For the giveDisplay the numbers in ascending order.

- Display only the numbers that are even.
- Calculate the factorial of each number.
- Display the sum of all factorial values.
- Display the largest factorial value.
- Display the final analysis in a well-organized format.

- In list, the program should:

5. Bank Transaction system - user-defined module, standard-library module, functions, scope.

A bank wants to generate transaction records for customers.

The program should allow:

- Deposit
- Withdrawal
- Balance inquiry
- Transfer

Create a reusable module containing the transaction-related operations.

The main program should additionally generate:

- Unique transaction ID
- Transaction date
- Transaction time

The system must reject invalid transactions such as negative amounts and withdrawals exceeding the available balance.

6. Employee Bonus & Performance Tracker — Scope

A company wants to maintain an employee's performance information and calculate a bonus based on performance.

Write a program that accepts an employee's name and basic salary. The program should maintain a company-wide bonus percentage, while each employee may have a different performance score.

The program must:

- Store the company bonus percentage so that it is available to the salary-calculation function.
- Calculate the employee's bonus based on the performance score.
- Display the employee's basic salary, bonus percentage, bonus amount, and final salary.
- Allow the bonus percentage to be changed by the appropriate operation and reflect the updated value in subsequent calculations.
- Create an internal/helper operation for calculating the bonus and demonstrate how it accesses values defined outside its own local scope.

- Ensure that changing a local variable with the same name as the company bonus percentage does not unintentionally change the company-wide value.

Example:

Company Bonus Percentage: 10%

Employee Name: Rahul

Basic Salary: ₹50,000

Performance Score: 85

Bonus Percentage: 10%

Bonus Amount: ₹5,000

Final Salary: ₹55,000

After changing the company bonus percentage:

New Company Bonus Percentage: 15%

Employee Name: Neha

Basic Salary: ₹60,000

Performance Score: 90

Bonus Percentage: 15%

Bonus Amount: ₹9,000

Final Salary: ₹69,000